

This is CS50

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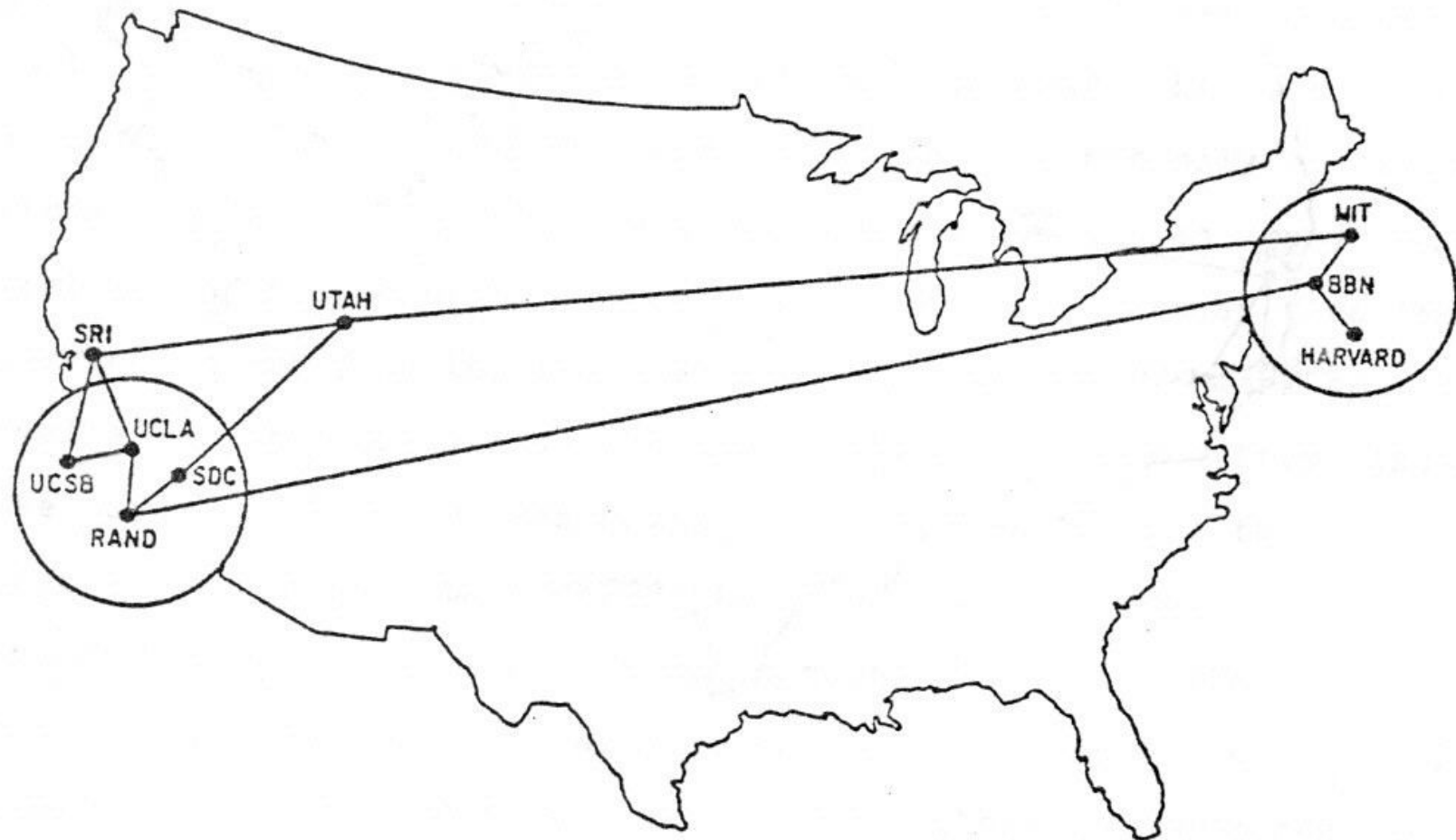
- Today's snack, Rice Krispies Treats. 😊
- No CS50 Lunch this week. The last one will be next week!
- Do **say hi or ask questions** during break or after class!
 - What do today's binary bulbs spell?

This is CS50

internet



The ARPANET in December 1969



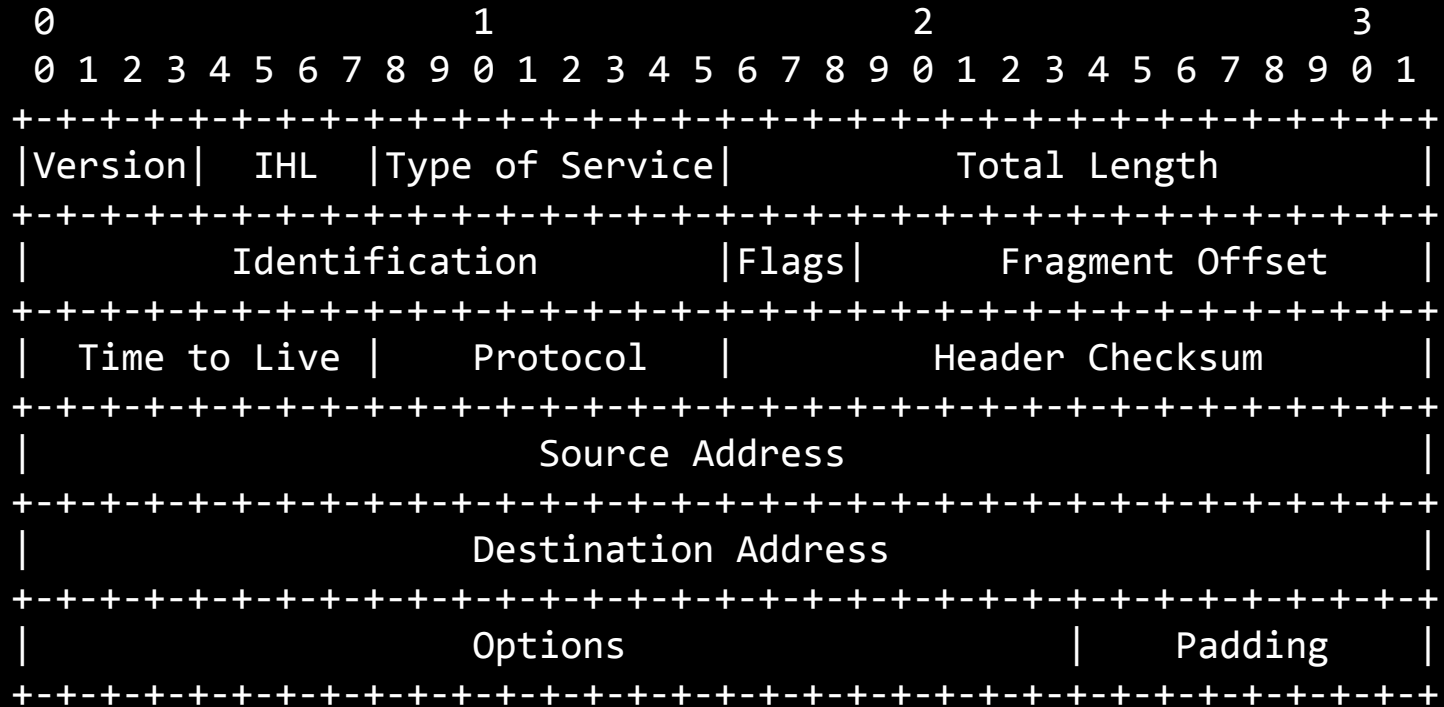
routers



TCP/IP

IP

###.



Example Internet Datagram Header

TCP

ports

80 HTTP

443 HTTPS

...



TCP Header Format

DNS

Domain Name	IP Address

Fully Qualified Domain Name	IP Address

DHCP

HTTP

HTTPS

<https://www.example.com>

<https://www.example.com/>

<https://www.example.com/path>

<https://www.example.com/file.html>

<https://www.example.com/folder/>

<https://www.example.com/folder/file.html>

<https://www.example.com/folder/file.html>

<https://www.example.com/folder/file.html>

<https://www.example.com/folder/file.html>

<https://www.example.com/folder/file.html>

<https://www.example.com/folder/file.html>

GET

POST

...

GET / HTTP/2

Host: www.harvard.edu

...

HTTP/2 200

Content-Type: text/html

...

curl

HTTP/2 301

Location: <https://www.harvard.edu/>

...

HTTP/2 404

Content-Type: text/html

...

200 OK
301 Moved Permanently
302 Found
304 Not Modified
307 Temporary Redirect
401 Unauthorized
403 Forbidden
404 Not Found
418 I'm a Teapot
500 Internal Server Error
503 Service Unavailable
...

GET / HTTP/1.1

Host: safetyschool.org

...

developer tools

HTML

tags

attributes

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
  <head>
```

```
    <title>
```

```
      hello, title
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    </title>
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  </head>
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  <body>
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    hello, body
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  </body>
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</html>
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http-server

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    </title>
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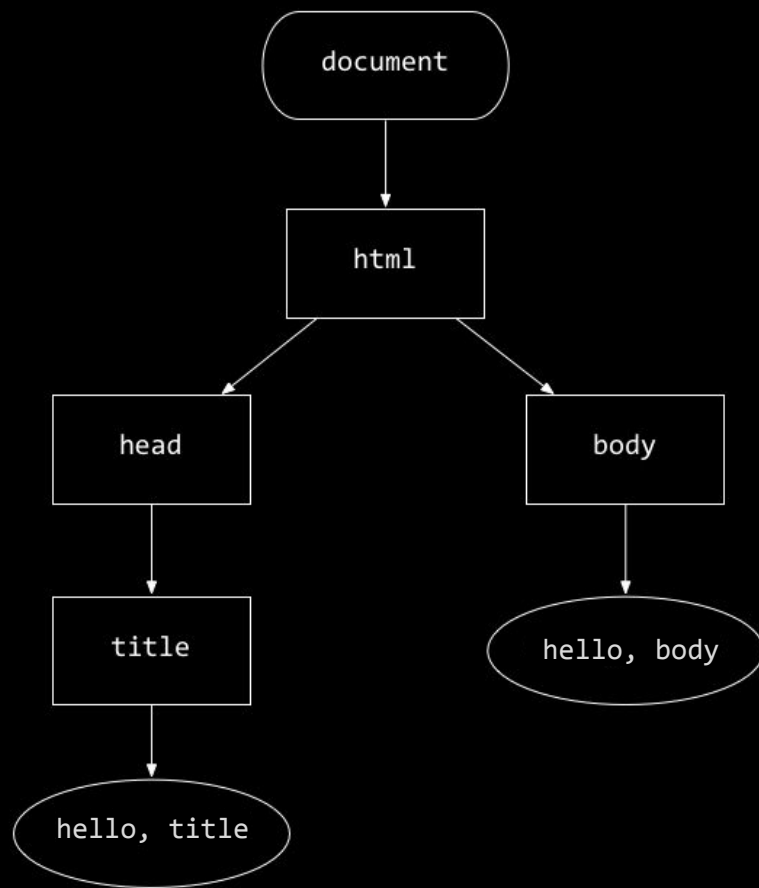
```
  </head>
```

```
  <body>
```

```
    hello, body
```

```
  </body>
```

```
</html>
```



<https://www.example.com>

<https://www.example.com/path?key=value>

<https://www.example.com/path?key=value&key=value>

regular expressions

regexes

developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Regular_expressions

.	any single character (except line terminators)
*	zero or more times
+	one or more times
?	0 or 1 time
{n}	n occurrences
{n,m}	at least n occurrences, at most m occurrences
...	

`[0123456789]` any one of the enclosed characters

`[0-9]` any one of the range of characters

`\d` any digit

`\D` any character that is not a digit

...

^[a-zA-Z0-9. !#\$%&' *+ \/= ?^_` { | } ~ -] + @[a-zA-Z0-9] (? : [a-zA-Z0-9 -] { 0 , 61 } [a-zA-Z0-9]) ? (? : \ . [a-zA-Z0-9] (? : [a-zA-Z0-9 -] { 0 , 61 } [a-zA-Z0-9]) ?) * \$

developer tools

validator.w3.org

CSS

properties

type selector

class selector

ID selector

attribute selector

...

```
<!DOCTYPE html>
```

```
<html lang="en">
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  <head>
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    <title>
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      hello, title
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    </title>
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  </head>
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  <body>
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    hello, body
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  </body>
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</html>
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<!DOCTYPE html>
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<html lang="en">
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    <title>
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      hello, title
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    </title>
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    hello, body
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  </body>
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</html>
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<!DOCTYPE html>
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<html lang="en">
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  <head>
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```
    <style>
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    </style>
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    <title>
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      hello, title
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    </title>
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  </head>
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  <body>
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    hello, body
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  </body>
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</html>
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<!DOCTYPE html>
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<html lang="en">
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  <head>
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    <style>
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    </style>
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    <title>
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      hello, title
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    </title>
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  </head>
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  <body>
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    hello, body
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  </body>
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</html>
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<!DOCTYPE html>
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<html lang="en">
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  <head>
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    <title>
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      hello, title
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    </title>
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  </head>
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  <body>
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    hello, body
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  </body>
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</html>
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```
<!DOCTYPE html>
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<html lang="en">
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  <head>
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```
    <link href="styles.css" rel="stylesheet">
```

```
    <title>
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```
      hello, title
```

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    </title>
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  </head>
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  <body>
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    hello, body
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  </body>
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</html>
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developer tools

frameworks

Bootstrap

getbootstrap.com

JavaScript


```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
  <head>
```

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    <title>
```

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      hello, title
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```
    </title>
```

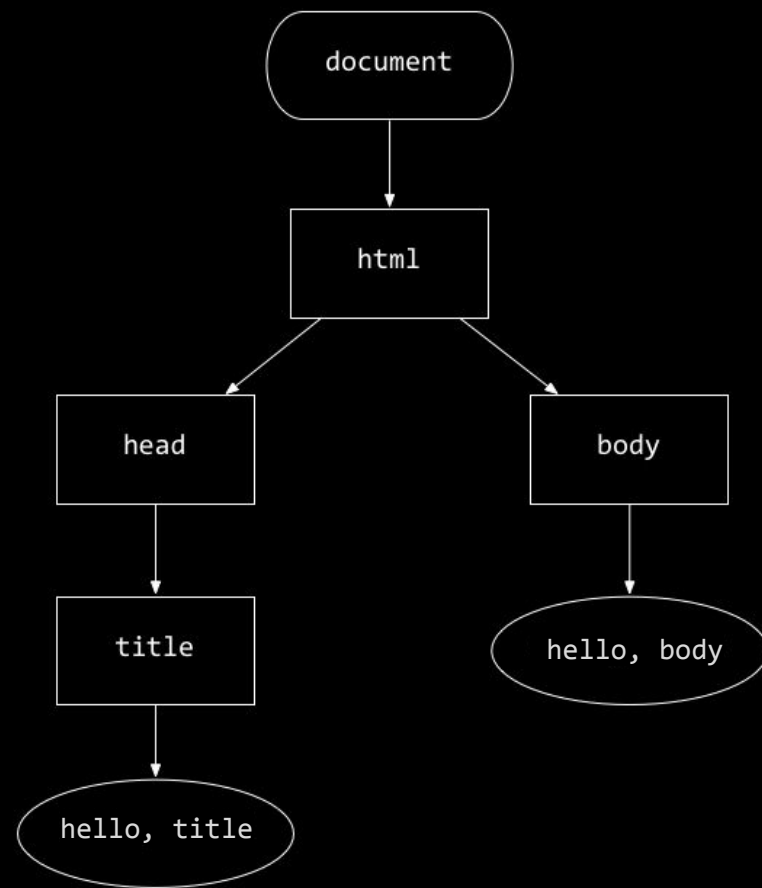
```
  </head>
```

```
  <body>
```

```
    hello, body
```

```
  </body>
```

```
</html>
```

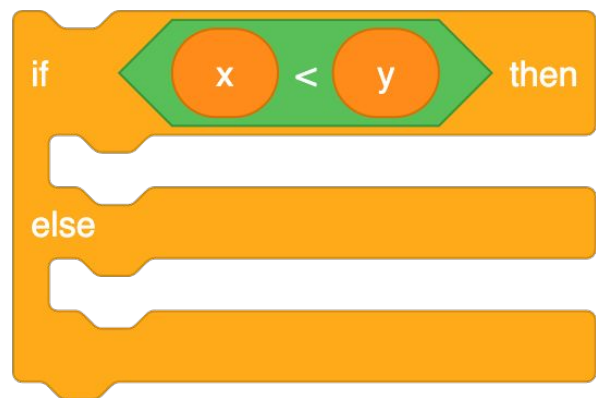


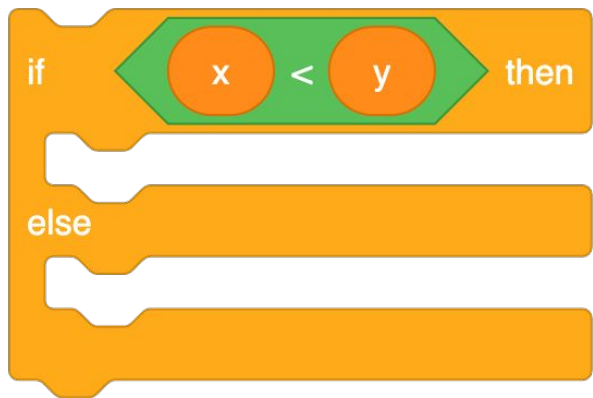
conditionals





```
if (x < y)
{
}
}
```

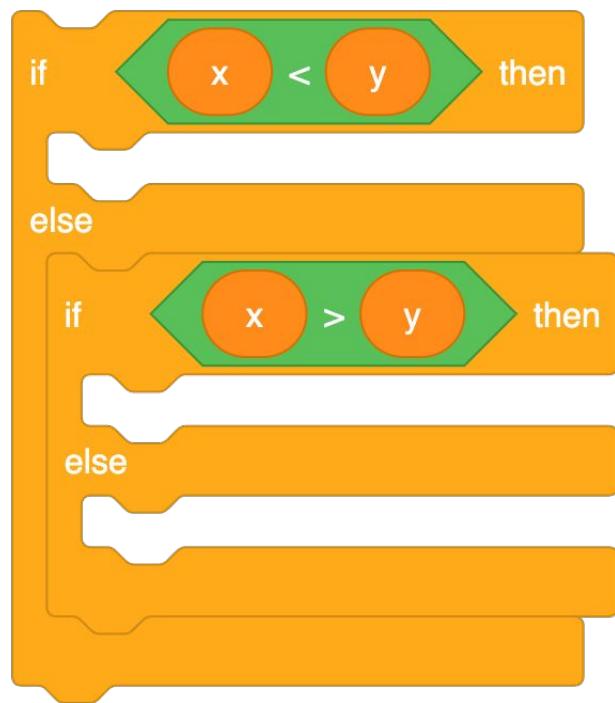


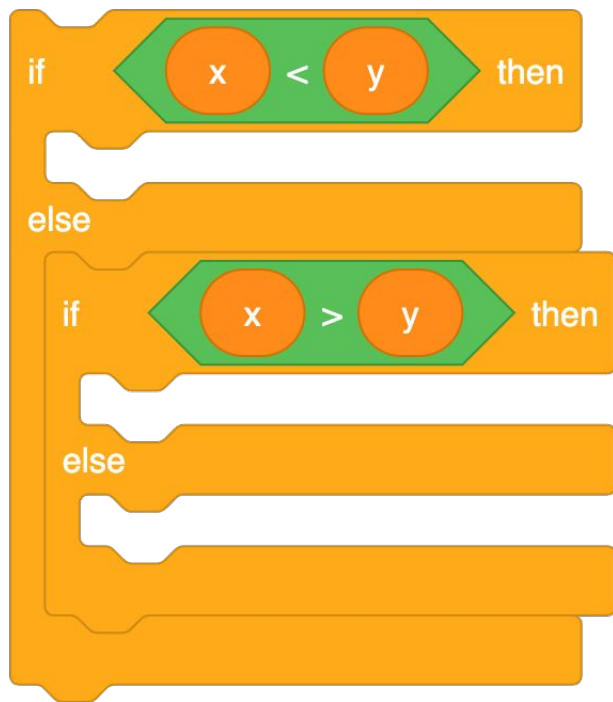


```
if (x < y)
{

}
else
{

}
```





```
if (x < y)  
{
```

```
}
```

```
else if (x > y)  
{
```

```
}
```

```
else  
{
```

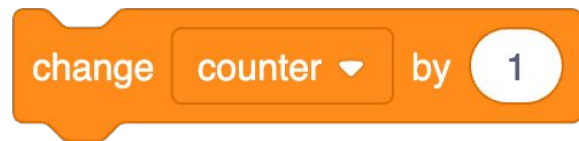
```
}
```

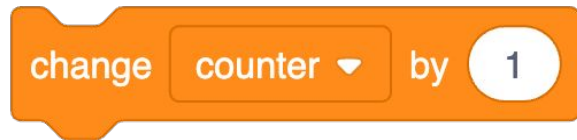

variables



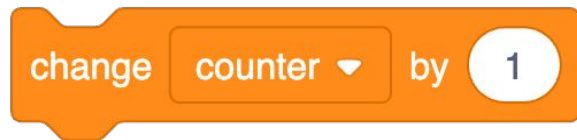


```
let counter = 0;
```

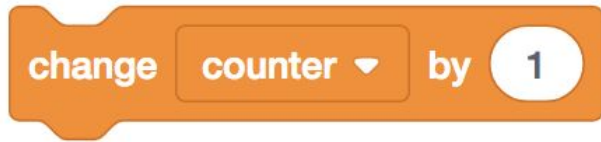




```
counter = counter + 1;
```

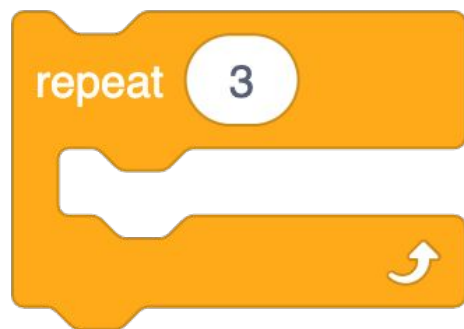


```
counter += 1;
```



```
counter++;
```

loops





```
for (let i = 0; i < 3; i++)  
{  
  
}
```





```
while (true)
{
}
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
  <head>
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    <title>
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      hello, title
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<!DOCTYPE html>
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  <head>
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    <script>
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    <title>
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<!DOCTYPE html>
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<html lang="en">
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  <head>
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```
    <script src="scripts.js"></script>
```

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    <title>
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      hello, title
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    </title>
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  </head>
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    <script>
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    <script src="scripts.js"></script>
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  </body>
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```
</html>
```

blur
change
click
drag
focus
keyup
load
mousedown
mouseover
mouseup
submit
touchmove
unload
...

This is CS50

ruse

 Helen Cho

 Jonny Zhang



Abel



 Sophie Edouard



 Moni Radev



ehreteab



 Joshua Ng



 Montague Mawere